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Unemployment Risk

Fluctuations in the risk of a large increase in unemployment are examined. The analysis compares medium-term risks—that is, risks at a 3-year horizon—to those over a 1-year horizon. The primary approach involves quantile regressions. Robustness exercises examine risks using a logistic regression to model the probability of a large increase in the unemployment rate. U.S. experience reveals an elevated risk of large increases in unemployment over the medium term when credit growth is high and when the unemployment rate is low. Near-term risks to unemployment are closely tied to changes in corporate bond spreads or the slope of the yield curve, consistent with research on recession prediction, but these factors do not play a sizable role in medium-term risks. These results highlight the value of considering different near- and medium-term risk factors.

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> "[A] central bank seeking to maximize its probability of achieving its goals is driven, I believe, to a risk-management approach to policy. By this I mean that policymakers need to consider not only the most likely future path for the economy but also the distribution of possible outcomes about that path."
>
> Federal Reserve Board Chairman Alan Greenspan, 29 August 2003.

Policymakers have long been concerned about risks to the achievement of their objectives. For example, participants in the Federal Open Market Committee (FOMC) assess the balance and magnitude of risks to their individual projections of inflation and unemployment. Similarly, central banks have used fan charts to summarize their expectation for the distribution of possible outcomes, including assessments of unusually large risks (e.g., the width of the fan charts) and the balance of risks (e.g.,

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skew relative to expected outcomes).[1] The International Monetary Fund (IMF) has increasingly focused on measures of risks to economic activity in its *Global Financial Stability Report*.

Until the impact of the COVID-19 pandemic in March 2020, the unemployment rate in the United States had fallen to a level at the low end of its experience, and some had questioned whether the late stages of an economic expansion might sow the seeds of the expansion's end. For example, the FOMC minutes of March 2018 noted, "A few participants expressed concern that a lengthy period in which the economy operates beyond potential and financial conditions remain highly accommodative could, over time, pose risks to financial stability."

I examine changes over time in the risk of large increases in the U.S. unemployment rate (defined more precisely below) using quantile regressions, as a method to gauge time variation in the magnitude for a conditional quantile of the change in the unemployment rate, and logit regressions to predict the probability of a large change in the unemployment rate of fixed size. Quantile regressions reveal an elevated risk of a large increase in unemployment over a 3-year horizon when the unemployment rate is low. In contrast, the level of the unemployment rate has little effect on the risk of a large increase in unemployment over the next year from a logit regression—that is, low levels of the unemployment rate point to some elevated medium-term risk to future unemployment, but not to elevated near-term risk. Importantly, these relationships gauge reduced-form correlations, and empirical work and structural modeling to examine the causal mechanisms at play are important (as in, for example, Duarte and Adrian 2017).

The empirical analysis considers a broader set of macro-economic and financial variables. Previous research has highlighted the growth of credit relative to GDP as a factor that predates financial crises, and the Basel Committee has emphasized this variable in its guidance regarding the countercyclical capital buffer (BCBS 2010).[2] The results herein suggest credit growth (relative to GDP growth) has been a correlate of unemployment risk at medium-term horizons. For example, rapid credit growth implied elevated risk of a large increase in unemployment in the late 1980s and mid-2000s, while muted advances in credit implied a low amount of unemployment risk in the early-to-late 1990s and since 2011.

The analysis also considers inflation as a risk factor, as high inflation predated recessions in the United States prior to recent decades (Romer and Romer 1989). Quantile regressions suggest that inflation has been a correlate of unemployment risk at the 3-year horizon in the United States since the 1960s—primarily in the 1970s and early 1980s.

1. For example, the Monetary Policy Committee of the Bank of England includes such assessments in its *Inflation Report*.

2. Schularick and Taylor (2012), among others, find that credit growth has some power to predict financial crises. Kiley (2021) examines this conclusion and the empirical approach, highlighting issues related to empirical fit and the role for other risk factors, including growth in equity prices and house prices and the current account deficit.

Financial variables have been emphasized as predictors of near-term economic activity or recessions. For example, the corporate bond spread (as measured by the difference between the yields on a Baa-rated corporate bond and the 10-year Treasury security) and the term spread (as measured by the difference between the yield on the 10-year Treasury security and the federal funds rate) have been found to predict recessions.[3] The analysis herein shows that these variables are not significant correlates of medium-term unemployment risk according to quantile or logit regressions. In contrast, these factors are associated with risks to unemployment at a 1-year horizon according to quantile and logit regressions.[4]

The analysis of unemployment risk is related to the literature on predicting recessions. U.S. recessions are always accompanied by outsized increases in the unemployment rate, and hence the risk of a large increase in the unemployment rate is related to the risk of a recession. In addition to drawing potential predictors from this research (the term spread and the corporate bond spread), the recession prediction literature motivates consideration of logit models of the probability of a fixed-size increase in the unemployment rate to complement the quantile-regression approach. Specifically, I define large increases in the unemployment rate as increases that exceed the unconditional 80th percentile of the change in the unemployment rate over four quarters (which equals $\frac{3}{4}$ percentage point) or over 12 quarters (which equals 1.9 percentage points). The analysis considers the probability of such large increases in the unemployment rate as predicted by logit regressions. The results from this logit regression approach are similar to those from quantile regressions: at a near term horizon, the term spread and corporate bond spread are predictors of this measure of unemployment risk; at the 3-year horizon, credit growth, inflation, and the level of the unemployment rate are more important.

The analysis herein expands previous analyses of GDP at Risk and recession risk along a number of important dimensions. First, the focus is on unemployment, as the unemployment rate is arguably a better gauge of the cyclical state of the U.S. economy than the percent change in GDP (e.g., Basistha and Startz 2008). Results examining unemployment risk consider the robustness of findings using real GDP to alternative measures of economic activity. Second, inflation is included as a risk factor, which seems critical given the role of high inflation in driving shifts toward a more restrictive monetary stance and subsequent U.S. recessions (e.g., Romer and Romer, 1989).

3. These variables are used to predict economic activity or the probability of a recession in, for example, Lopez-Salido, Stein, and Zakrajsek (2017) and Engstrom and Sharpe (2018). Similar sets of variables are considered in Faust et al (2013) and Favara et al (2016).

4. Adrian, Boyarchenko, and Giannone (2019) emphasize the National Financial Conditions Index (NFCI) as a measure of financial conditions linked to tail outcomes at a 1-year horizon. The appendix to the final working paper version of this research (Kiley 2018) reports results for the NFCI (which is available for a shorter sample than the corporate bond and term spreads used herein). The results are broadly similar to those using the corporate bond spread and term spread (and to the 1-year horizon results in Adrian, Boyarchenko, and Giannone 2019). Specifically, an increase in the NFCI is associated with an increase in the adverse (high) tail for changes in the unemployment rate at a 1-year horizon, but not at a 3-year horizon (as measured by quantile regressions)—similar to the results using the corporate bond spread. At a 3-year horizon, the more rapid growth in credit over the previous 4 years is associated with an increase in the adverse tail outcome for the change in the unemployment rate.

Third, the analysis explicitly considers unemployment risk at different horizons (one year and three year), as joint consideration of these two horizons brings together the near-term focus common in the literature on recession prediction (e.g., Engstrom and Sharpe 2018) and the medium-term focus common in financial stability monitoring (e.g., IMF 2017). Fourth, the analysis considers both the quantile approach to defining risk in economic activity, as in Adrian, Boyarchenko, and Giannone (2019), and an alternative definition of unemployment risk based on a fixed-sized increase in the unemployment rate examined through a logit regression approach, as in the recession prediction literature.

Section 1 discusses the framework for assessing unemployment risk, including the definition of unemployment risk in terms of quantiles of the unemployment distribution, the key aspects of the quantile and logistic regression approaches, and the data used in this study. Section 2 considers the relationship between unemployment risk, the corporate bond spread, the term spread, nonfinancial private sector borrowing, and inflation. Section 3 concludes.

1. DATA AND FRAMEWORK

1.1 Data

The analysis focuses on the civilian unemployment rate for the noninstitutional population in the United States.[5] The unemployment rate rises sharply during recessions and declines during expansions, with the pace of declines during expansions generally more moderate than the pace of increases during recessions. This pattern is suggestive of some type of asymmetry in the distribution of changes in unemployment.

The factors that may influence the risks associated with the outlook for unemployment are drawn from the recent literature on factors that may presage financial instability as well as the larger (as well as older) literature on business cycles. For example, Cecchetti and Li (2008) and Adrian, Boyarchenko, and Giannone (2019) consider (alternative subsets of) financial variables and credit as potential predictors of quantiles of economic activity as measured by the percent change in GDP (i.e., GDP at Risk), including measures of credit and corporate bond spreads. Schularick and Taylor (2012) and Kiley (2021) show that multiyear averages of the change in credit relative to GDP appear to predict financial crises.[6] Recessions in the United

5. The initial working paper version of this research (Kiley 2018) also considered the unemployment rate for men aged 25–54. This subset of the population has historically had a more stable attachment to the labor force and hence the unemployment rate in this group has been emphasized in some research. Results were largely identical to those reported for the overall unemployment rate and hence readers are referred to Kiley (2018) for that analysis.

6. Alternative approaches to credit relative to GDP include consideration of the percent change over a shorter period or inclusion of multiple lags of such shorter-period percent changes. Analysis considering only a 1-year percent change (not reported) showed no relationship between such a change and unemployment risk. Given the sample sizes and focus on rare events (tail risks), specifications with many lags are likely to prove noisy and are not considered on *a priori* grounds. Note that Schularick and Taylor (2012) consider multiple lags in a much larger data set, but focus on the sum of such lags, and Kiley (2021) directly focuses on a multiyear change.

States have been shown to be preceded, albeit with relative short lags, by increases in credit spreads or declines in the term spread (e.g., Faust et al. 2013, Favara et al. 2016, Engstrom and Sharpe 2018). In addition, U.S. recessions—at least prior to 2000—tended to be associated with clear shifts in monetary policy to bring inflation down from undesirably high levels (Romer and Romer 1989).

In light of these earlier studies, the analysis examines the relationship between the unemployment rate and the change in the spread between a Baa-rated corporate bond and the 10-year Treasury yield, the term spread as measured by the difference between the yield on a 10-year Treasury security and the federal funds rate, the (log) change in the ratio of private-sector nonfinancial borrowing to nominal GDP over the previous four years (expressed at an annual rate), and the four-quarter percent change in the personal consumption expenditures price index.

1.2 Measuring Unemployment Risk

I define unemployment risk as the risk of a large increase in the unemployment rate. I focus on the change in the unemployment rate, rather than its level, for two reasons. First, this focus allows clear comparison with recent work on risks to the growth rate of real GDP, as the growth rate of real GDP is tied to the change in the unemployment rate by Okun's law. Second, focusing on the change in the unemployment rate will ensure a close connection between unemployment risk and recession risk and hence facilitate unifying perspectives from the literature on Growth at Risk and recession prediction. Note, however, that this focus is not restrictive, as the level of the unemployment rate will be among the predictors used and it is straightforward to translate statements regarding the risk of changes to levels.

Given the focus on the risk of a large increase in the unemployment rate, I consider two precise definitions of unemployment risk.

Unemployment Risk I: 80th percentile conditional on current information. The first definition of unemployment risk is the *size of the 80th percentile of the distribution of changes in the unemployment rate over horizon h at time t*, $Q_{0.80}^{\Delta U}(t+h)$. Denoting the cumulative distribution function of the change in the unemployment rate over horizon h conditional on time t information $I(t)$ as $F(\Delta U(t+h)|I(t))$, the 80th conditional percentile is {\fontsize{9}{11}

$$Q_{0.80}^{\Delta U}(t+h) = F^{-1}(0.80|I(t)) = \inf\{\Delta U(t+h) : F(\Delta U(t+h)|I(t)) \geq 0.80\}. \quad (1)$$

In words, the 80th conditional percentile of the change in the unemployment rate over horizon h is the smallest value of the change in the unemployment rate over that horizon such that there is an 80% (or greater) probability that the change in the unemployment rate will be less than the value. The definition holds fixed the probability of the event being considered—that is, the event has a fixed probability, and changes in unemployment risk according to this definition are measured in units of the size of the change in the unemployment rate. For example, unemployment risk at time t over horizon h may equal 3 percentage points—a very large increase in the unemployment rate to encompass 80% of the time t distribution of potential outcomes. Conversely, unemployment risk at period t over horizon h could equal $\frac{1}{4}$ percentage point—a very small increase in the unemployment rate to encompass 80% of potential outcomes.

The 80th conditional percentile is chosen as a compromise statistic that relates closely to a standard risk metric—a recession—while also lying close to the more extreme tail scenarios, such as the 90th or 95th (or 10th or 5th) percentile emphasized in Growth at Risk work. The next subsection illustrates the coherence between the 80th percentile and recessions in the United States, and the working paper version of this research (Kiley 2018) focused on quantiles more deeply in the tail, as in the Growth at Risk literature. Note that more extreme percentiles may be more difficult to estimate—that is, experience deep in the tails of distributions are rare and hence relationships in such tails are challenging to estimate.

To examine changes over time in $Q^{\Delta U}_{0.80}(t+h)$, the following equation, linking the potential risk factors discussed above to unemployment risk, is estimated using quantile regression:

$$\Delta U(t+h) = a_0 + a_1 U(t) + a_2 \Delta_4 p(t) + a_3 \Delta_{16}\left(\frac{c(t)}{y(t)}\right) + a_4\big(r_{bbb}(t) - r_{tsy}(t)\big) + a_5\left(r_{tsy}(t) - r_{ffr}(t)\right). \tag{2}$$

In equation (2), $\Delta_4 p(t)$ is the four-quarter change in (the natural logarithm of) the personal-consumption expenditures price index up through the current quarter, $\Delta_{16}(\frac{c(t)}{y(t)})$ is the change over the past 16 quarters (4 years) in (the natural logarithm of) nonfinancial private-sector credit outstanding divided by (nominal) GDP, $(r_{bbb}(t) - r_{tsy}(t))$ is the spread between yields on a BBB-rated corporate bond and a 10-year Treasury security, and $(r_{tsy}(t) - r_{ffr}(t))$ is the spread between the yield on a 10-year Treasury security and the federal funds rate. Based on conventional wisdom and historical narratives, high values of each of the first three variables is expected to increase projections of the unemployment rate and thereby raise unemployment risk, whereas low values of the term spread are expected to be associated with higher unemployment risk.

Before turning to results, a bit more review of quantile regressions may help some readers. While quantile regressions are less widely used in macroeconomics than least squares, the GDP at Risk literature has used the approach extensively (e.g., Cecchetti and Li 2008, IMF 2017, Adrian et al. 2018, Adrian, Boyarchenko, and Giannone 2019). Moreover, the intuition is straightforward: quantile regression simply weights errors in the projections more heavily for errors near the quantile of interest and less heavily for errors distant from the quantile of interest. To see this more formally, define the error terms consistent with equation (2) as $e(t)$ and note that quantile regression for a given quantile q minimizes the loss function

$$L = \sum_{t=1}^{T} qe(t)I(e(t) > 0) + (q-1)e(t)I(e(t) < 0),$$

where $I(.)$ is the indicator function (i.e., $I(e(t) < 0)$ equals 1 when $e(t) < 0$. For high quantiles (q above 0.5), positive errors receive larger weight than a negative error. Alternatively, the 50th percentile quantile regression—the median regression—finds the coefficients that minimize the least absolute deviation of the errors from the projection (rather than least squared deviation in ordinary least squares regression). This approach places relatively more weight on deviations close to the center of the error distribution (e.g., close to the estimated median) than least squares, as absolute deviations are relatively smaller for larger errors than are squared errors. (For a formal discussion of quantile regression, see Koenker and Hallock 2001). This intuition highlights why uncertainty regarding tail relationships (q far from 0.5) is challenging—such relationships are uncovered by weighting observations in the tails heavily, but such observations are by definition somewhat rare.

Some additional intuition regarding quantile regressions can be gleaned from scatterplots of the data and associated least-squares and quantile regression lines. Figure 1 presents bivariate scatterplots of the change in the unemployment rate from period t to period $t + 12$ (the medium-term horizon) against each variable that enters equation (2). In each panel, dots represent pairs of data in a given time period (over the sample 1965:Q1 to 2019:Q14). The solid line is a least squares regression, and the dashed line is the 80th quantile regression. For example, in the lower left panel involving the term spread, the regression lines are roughly parallel, indicating that the coefficient on the term spread is roughly equal in the two regressions. This is apparent from the cloud of data in the scatterplot, which has a similar span across the y-axis for the range across the x-axis. In contrast, in the middle left panel involving the percent change in credit relative to GDP over the previous 4 years, the quantile regression line has a notably steeper slope, indicating that the coefficient on credit/GDP is larger in the quantile regression. This relationship reflects the pattern in the cloud of data: for high values of growth in credit/GDP, there is a mass of values of the increase in the unemployment rate at high values as well as some at lower values, whereas such values do not appear when growth in credit/GDP is low.

The bivariate plots provide intuition, but the core of the analysis will focus on multivariate relationships.

Unemployment Risk II: Probability that the change in unemployment exceeds its unconditional 80th percentile based on current information. The second definition of unemployment risk is the *probability that the change in the unemployment rate over horizon h, condition on period t information, exceeds the 80*th *percentile of the unconditional distribution of changes in the unemployment rate*, $P^{\Delta U}_{0.80}(t + h)$. Denoting

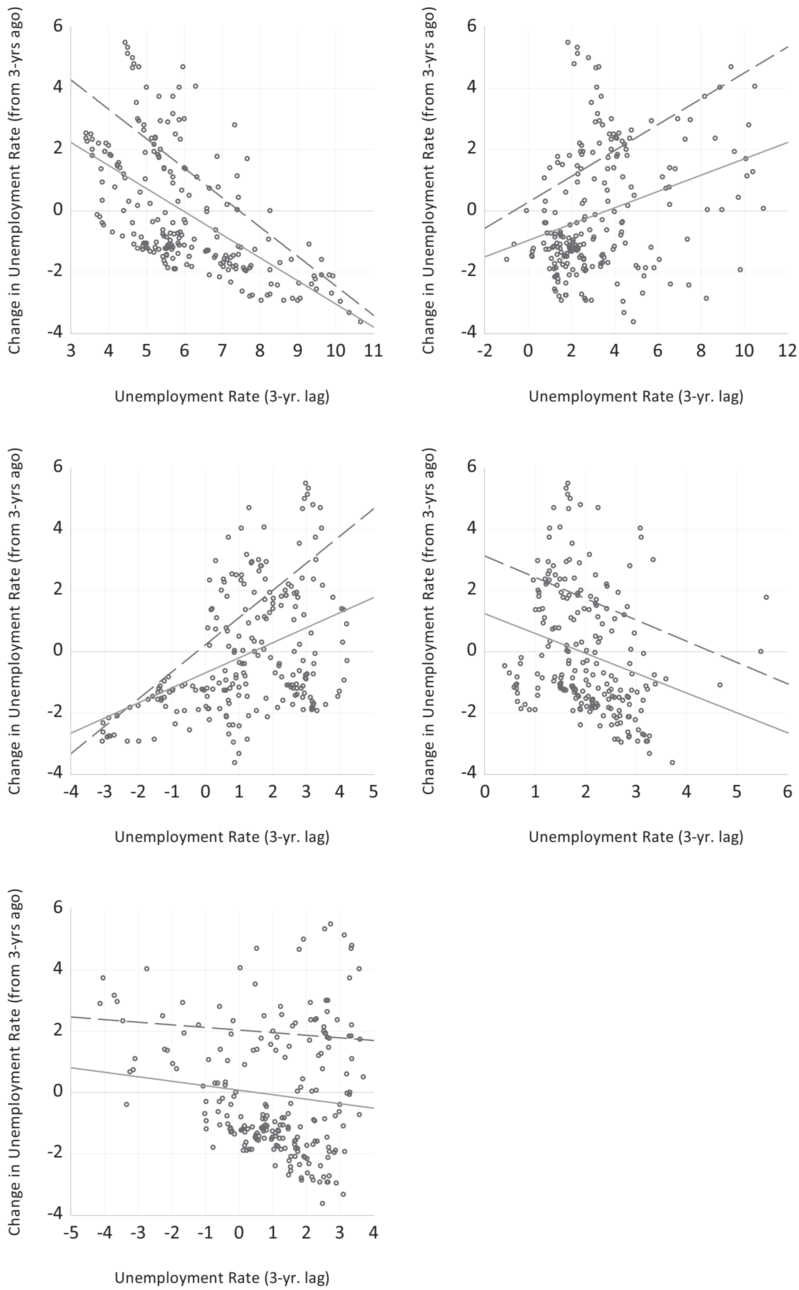

Fig 1. Bivariate Scatterplots of the Data and Associated Least Squares and 80th Quantile Regression Lines.

Notes: In each panel, dots represent pairs of data in a given time period (over the sample 1965:Q1 to 2019:Q14). The solid line is a least squares regression, and the dashed line is the 80th quantile regression. For example, in the lower left panel involving the term spread, the regression lines are roughly parallel, indicating that the coefficient on the term spread is roughly equal in the two regressions. In contrast, in the middle left panel, the quantile regression line has a notably steeper slope, indicating that the coefficient on credit/GDP is larger in the quantile regression. Based on author's calculations using data retrieved from FRED, Federal Reserve Bank of St. Louis.

the unconditional cumulative distribution function of change in the unemployment rate over horizon h as $F(\Delta U(t+h))$, $P_{0.80}^{\Delta U}(t+h)$ is

$$P_{0.80}^{\Delta U}(t+h) = \ Prob\left\{\Delta U(t+h) \geq F^{-1}(0.80)\right\}. \tag{3}$$

The definition holds fixed the size of the increase in unemployment, at the 80th percentile of the unconditional distribution $F^{-1}(0.80)$ (where the absence of a reference to period t refers to the unconditional distribution). Time variation in unemployment risk is measured in probability units—that is, how likely is an increase in the unemployment rate equal to or greater than its historical 80th percentile over the horizon of interest.

Defining unemployment risk as the 80th percentile of the change in the unemployment rate links closely to recession risk. At a horizon of four quarters, the 80th percentile of the change in the unemployment rate equals $\frac{3}{4}$ percentage point; at 12 quarters, it equals 1.9 percentage points. Changes in the unemployment rate at these horizons of this magnitude are always associated with recessions.

In light of the link to recession risk, the empirical approach to gauging time variation in this measure of unemployment risk is drawn from the literature on predicting recessions. In particular, I estimate a logistic regression of the following form:

$$\begin{aligned} Prob\left\{\Delta U(t+h) \geq F^{-1}(0.80)\right\} &= G\left(a_0 + a_1 U(t) + a_2 \Delta_4 p(t)\right. \\ &\left. + a_3 \Delta_{16}\left(\frac{c(t)}{y(t)}\right) + a_4\left(r_{bbb}(t) - r_{tsy}(t)\right) + a_5\left(r_{tsy}(t) - r_{ffr}(t)\right)\right), \end{aligned} \tag{4}$$

where $G(X) = 1/(1+e^{-X})$ (the logistic function).

2. PREDICTORS OF UNEMPLOYMENT RISK

The analysis uses data from 1965:Q1 to 2019:Q4, with the start date determined by the availability of data on the spread between a BBB-rated corporate bond and the 10-year Treasury yield.

2.1 *Relationship of Conditional 80th Percentile Unemployment Risk, $Q_{0.80}^{\Delta U}(t+h)$, to Predictors*

Table 1 reports coefficient estimates for the quantile regressions, focusing on a horizon of 1 year (four quarters) and 3 years (12 quarters). Results for least-squares regressions are also presented, for comparison purposes.

The pattern of coefficients in the least-squares regression at the 1-year horizon is useful to understand subsequent results relative to previous work. Four of the five predictors are estimated to have statistically significant links to the change in the unemployment rate over the next year: the current level of the unemployment rate, the current inflation rate, the corporate bond spread, and the term spread. These results link directly to views common in the literature. The unemployment rate is mean

TABLE 1

REGRESSIONS FOR THE CHANGE IN THE UNEMPLOYMENT RATE AT ALTERNATIVE HORIZONS.

$$\Delta U(t+h) = a_0 + a_1 U(t) + a_2 \Delta_4 p(t) + a_3 \Delta_{16}\left(\frac{c(t)}{y(t)}\right) + a_4(r_{bbb}(t) - r_{tsy}(t)) + a_5(r_{tsy}(t) - r_{ffr}(t))$$

	$Q_{0.80}^{\Delta U}(t+h)$(Quantile Regression)Horizon h = 4 Quarters (1 Year)		$Q_{0.80}^{\Delta U}(t+h)$(Quantile Regression)Horizon h = 12 Quarters (3 years)		$E\{\Delta U(t+h)\}$(Least Squares)Horizon $h = 4$ Quarters (1 year)		$E\{\Delta U(t+h)\}$(Least Squares)Horizon $h =$ 12 Quarters (3 years)	
	Coefficient	90% Confidence Interval	Coefficient	90% Confidence Interval	Coefficient	90% Confidence Interval	Coefficient	90% Confidence Interval
a_1	**−0.12**	**(−0.44, 0.00)**	**−0.68**	**(−1.14, −0.33)**	**−0.48**	**(−0.43, −0.08)**	**−0.79**	**(−1.09, −0.50)**
a_2	**0.26**	**(0.05, 0.47)**	0.45	(−0.09, 0.69)	**0.15**	**(0.06, 0.28)**	**0.31**	**(0.03, 0.50)**
a_3	0.16	(−0.12, 0.30)	**0.39**	**(0.03, 0.69)**	0.07	(−0.07, 0.18)	0.22	(−0.01, 0.46)
a_4	**0.62**	**(0.00, 1.01)**	0.55	(−0.47, 1.10)	**0.57**	**(0.07, 0.91)**	0.49	(−0.27, 0.91)
a_5	−0.23	(−0.47, 0.06)	−0.10	(−0.65, 0.09)	**−0.23**	**(−0.35, −0.07)**	−0.20	(−0.65, 0.08)
R^2	0.38		0.43		0.25		0.61	

NOTES: Intercept (a_0) results not reported. Remaining coefficients refer to the equation at the top of the table. In this equation, the variables are
• Dependent variable $\Delta U(t+h)$: Change in U.S. civilian unemployment rate from period t to period $t+h$ (where h equals 4 or 12), from U.S. Bureau of Labor Statistics;
• Coefficient and variable a_1 : $U(t)$—U.S. civilian unemployment rate in period t, from U.S. Bureau of Labor Statistics;
• Coefficient and variable a_2 : $\Delta_4 p(t)$—PCE prices (percent change from four quarters earlier), from U.S. Bureau of Economic Analysis;
• Coefficient and variable a_3 : $\Delta_{16}(\frac{c(t)}{y(t)})$—Total nonfinancial credit divided by nominal GDP (percent change from 4 years earlier, expressed at an annual rate), from Bank for International Settlements (credit) and U.S. Bureau of Economic Analysis (nominal GDP);
• Coefficient and variable a_4 : $(r_{bbb}(t) - r_{tsy}(t))$—Difference between yields on Baa corporate bonds and 10-year Treasury security, from Moody's, Inc. and the U.S. Treasury;
• Coefficient and variable a_5 : $(r_{tsy}(t) - r_{ffr}(t))$—Difference between yield on 10-year Treasury security and the federal funds rate, from the U.S. Treasury and Federal Reserve Board.
All series retrieved from FRED, Federal Reserve Bank of St. Louis.
Estimation period is 1965Q1 to 2019Q4. In all cases, confidence intervals derived via a block-of-block bootstrap using resampling of the blocks of all variables in the equation (see Kilian and Lütkepohl, 2017, chapter 12), in which the block length equals 11. For quantile regressions, R^2 for quantile regressions is the measure from Koenker and Machado (1999). Bolded entries indicate statistical significance at the 10 percent level.

reverting, so a high value of the unemployment rate suggests a lower (more negative) change in the unemployment rate ($a_1 < 0$). High inflation has led to tight monetary policy and recession (Romer and Romer 1989), and hence high inflation suggests a larger change in the unemployment rate ($a_2 > 0$). An increase in the corporate bond spread often predates a weakening in economic activity (Faust et al. 2013, López-Salido, Stein, and Zakrajsek 2017), consistent with $a_4 > 0$. Conversely, a low value of the term spread often predates a recession (Engstrom and Sharpe 2018), consistent with $a_4 > 0$. Finally, the change in (the natural logarithm of) credit relative to GDP (over the previous four years) is not significant and the estimated coefficient is small in the least-squares regression of the change in the unemployment rate over the next year.

Several of these patterns change, or are more tenuous in the statistical sense, when examining the relationship of these predictors to the adverse tail outcome—$Q_{0.80}^{\Delta U}(t+4)$—via quantile regression. The role of mean reversion in the unemployment rate is much smaller (as the coefficient is only 0.12, a fraction of the least-squares value), while the role of inflation is similar in magnitude. Both the corporate bond spread and the term spread enter the quantile regression with coefficients similar in magnitude to the least-squares values, but only the corporate bond spread is statistically significant. This may not be surprising: the quantile regression involves a large number of variables (five) and it may be difficult to obtain strong results, as the focus is on the adverse tail and tail outcomes are infrequent and hence difficult to estimate.

Over the 3-year horizon, the quantile regression results are notably different—in the magnitude of coefficients and in their statistical significance. $Q_{0.80}^{\Delta U}(t+12)$ appears to be linked to the level of the unemployment rate (as the coefficient is near -0.7, approximately the least-squares value, and its 90% confidence interval excludes zero by a wide margin). The coefficient on credit growth is large (0.39)—much larger than the least-squares coefficient—and statistically significant. In contrast, other coefficients are not statistically significant at conventional levels. However, the scale of the coefficients remains similar to those at the 1-year horizon. In other words, statistical significance is tenuous; again, this may be expected given a focus on the tail and the limited number of observations to inform the quantile regression.[7]

Turning from a statistical to an economic description, the variation over time in risk can be understood better through a historical decomposition of unemployment risk $Q_{0.80}^{\Delta U}(t+h)$ implied by the estimated quantile regressions. Figure 2 summarizes the variation in this measure of the upper tail of unemployment risk as predicted by equation (2) for the 1-year horizon, computed as the coefficient multiplied by the demeaned variable (with the overall measure equaling the sum of these measures). This decomposition effectively normalizes the role of each variable by combining the variables movements with its coefficient. The visual presentation emphasizes three

7. In light of the possibility that the magnitude or statistical strength of links between the predictors and $Q_{0.80}^{\Delta U}(t+h)$ is masked by considering all predictors jointly, quantile regressions that include the level of the unemployment rate and each additional predictor individually, rather than jointly, were considered. In broad terms, the results are similar for each variable and hence not reported.

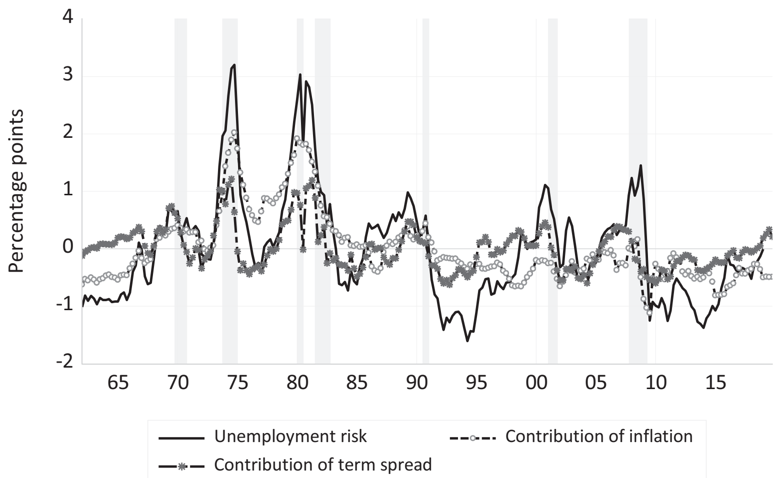

Fig 2. Unemployment Risk $Q_{0.80}^{\Delta U}(t+4)$ and Contributions of Explanatory Variables over Horizon of Four Quarters.

NOTES: The solid line represents unemployment risk as measured by the conditional 80th quantile consistent with the quantile regression in Table 1 at a horizon of four quarters (demeaned). The dashed lines with symbols are the contribution of the respective variable to this quantity, as measured by the coefficient from the quantile regression multiplied by the variable (demeaned). Based on author's calculations using data retrieved from FRED, Federal Reserve Bank of St. Louis. The shaded regions are periods identified as recessions by the National Bureau of Economic Research.

points. First, this measure of risk clearly rises near recessions—that is, in advance of recessions in the mid 1970s, 1980, 1990, 2000, and 2008. Second, the level of inflation is estimated to contribute sizably, especially in the 1970s and early 1980s. This finding is based on the entire sample but confirms the idea that increases in inflation have predated recessions, perhaps because monetary policy tightened to bring inflation down at such times (Romer and Romer 1989). Finally, the term spread generally signals an increase in risk somewhat before recessions, but this signal is not outsized relative to others in the figure.

The visual impression at the 3-year horizon is similar along some dimensions and different along others, as can be seen in Figure 3. As at the 1-year horizon, tail risk rises before and around recessions, and the inflation contribution appears large in the 1970s and early 1980s. However, the quantitative contributions estimated for other predictors are quite different. The contribution from the growth of credit relative to GDP is relatively sizable, as is the contribution of the level of the unemployment rate. In particular, rapid credit growth or low unemployment appear to increase notably the size of the adverse tail for the unemployment rate at a horizon of 3 years. For example, rapid credit growth in the 1980s signals, within the quantile regression of equation (2), a larger increase in the unemployment rate as measured by the conditional 80th percentile in the 3 years following 1987. The contribution of this variable rises to a percentage point in 1987, reflecting credit growth in excess of GDP growth of 4 percentage points per year over the previous 4 years, about $2\frac{1}{2}$ percentage points

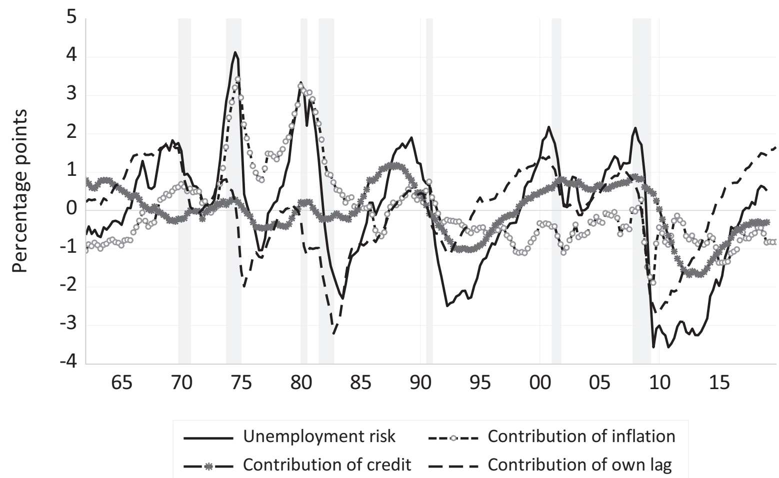

Fig 3. Unemployment Risk $Q^{\Delta U}_{0.80}(t+12)$ and Contributions of Explanatory Variables over Horizon of 12 Quarters.

NOTES: The solid line represents unemployment risk as measured by the conditional 80th quantile consistent with the quantile regression in Table 1 at a horizon of 12 quarters (demeaned). The dashed lines with symbols are the contribution of the respective variable to this quantity, as measured by the coefficient from the quantile regression multiplied by the variable (demeaned). Based on author's calculations using data retrieved from FRED, Federal Reserve Bank of St. Louis. The shaded regions are periods identified as recessions by the National Bureau of Economic Research.

above the variable's mean (i.e., 2.5 multiplied by the coefficient of about 0.4 yields a contribution of 1 percentage point). This finding is consistent with the recent literature on financial crises (e.g., Kiley 2021, Schularick and Taylor 2012)—that is, rapid credit growth predated the Savings and Loans Crisis in the United States and the subsequent U.S. recession.

2.2 Relationship of Probability of Change in Unemployment Rate Greater than Unconditional 80th Percentile, $P^{\Delta U}_{0.80}(t+h)$, to Predictors

The results for the quantile regressions can be compared to those associated with the alternative measure of unemployment risk—the probability that unemployment rises by an amount greater than its unconditional 80th percentile over either 1 or 3 years as predicted by the logit regression presented in equation (4).

Figure 4 presents the probability of such an increase from the regressions. At both horizons, it is clear that there is some predictive power, at least *ex post* (as these are full sample regressions)—the probabilities rise strongly in advance of recessions.

To gauge the importance of the variables, marginal effects (evaluated at sample means and scaled to correspond to a change of one standard deviation in the variable) are presented in Figure 5. The results both echo results in earlier literature and from the quantile regressions. At a 1-year horizon, the marginal effects of the term spread and the corporate bond spread are sizable and statistically significant: a low value of

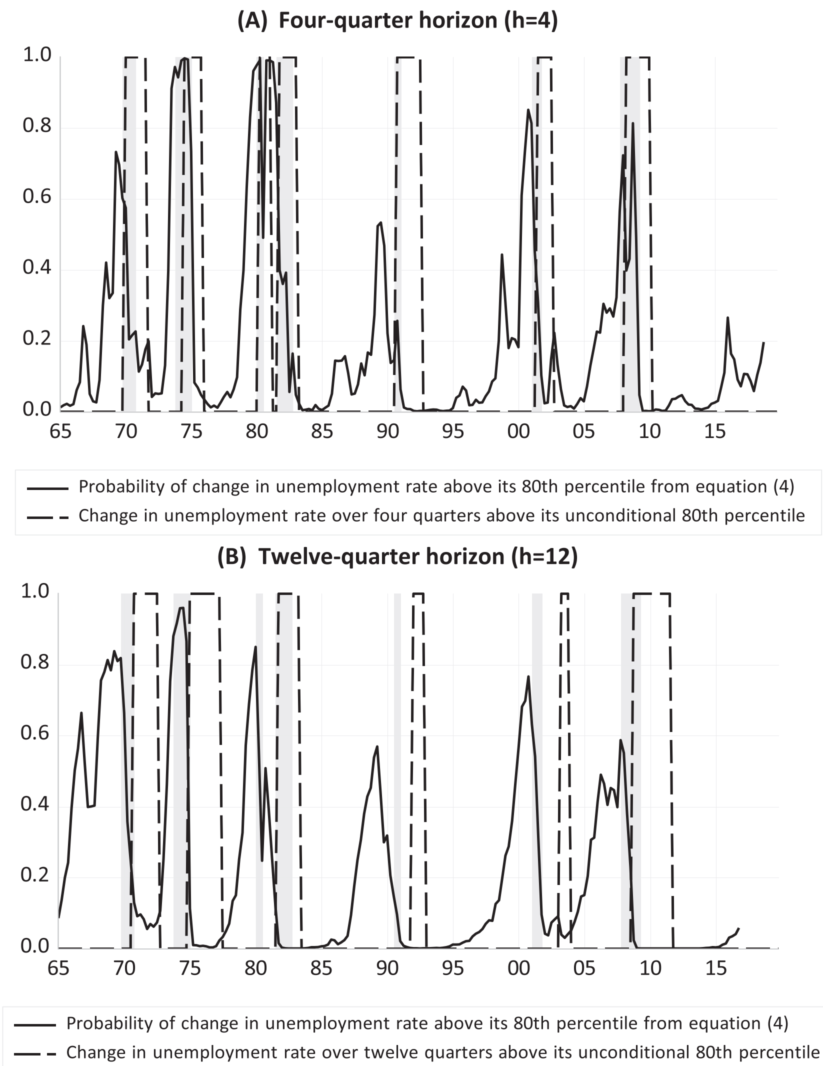

Fig 4. Unemployment Risk as Measured by the Probability of an Increase in the Unemployment Rate above its Unconditional 80th Percentile $P_{0.80}^{\Delta U}(t+h)$. (A) Four-Quarter Horizon ($h=4$). (B) Twelve-Quarter Horizon ($h=12$).

Notes: The series graphed as a dashed line equals 1 when the change in the unemployment rate from 4 or 12 quarters earlier exceeds its unconditional 80th percentile (which equal $\frac{3}{4}$ percentage point and 1.9 percentage point, respectively) and 0 otherwise. The series graphed as a solid line is the predicted probability of increases in the unemployment rate of these magnitudes at these horizons from equation (4). Based on author's calculations using data retrieved from FRED, Federal Reserve Bank of St. Louis. The shaded regions are periods identified as recessions by the National Bureau of Economic Research.

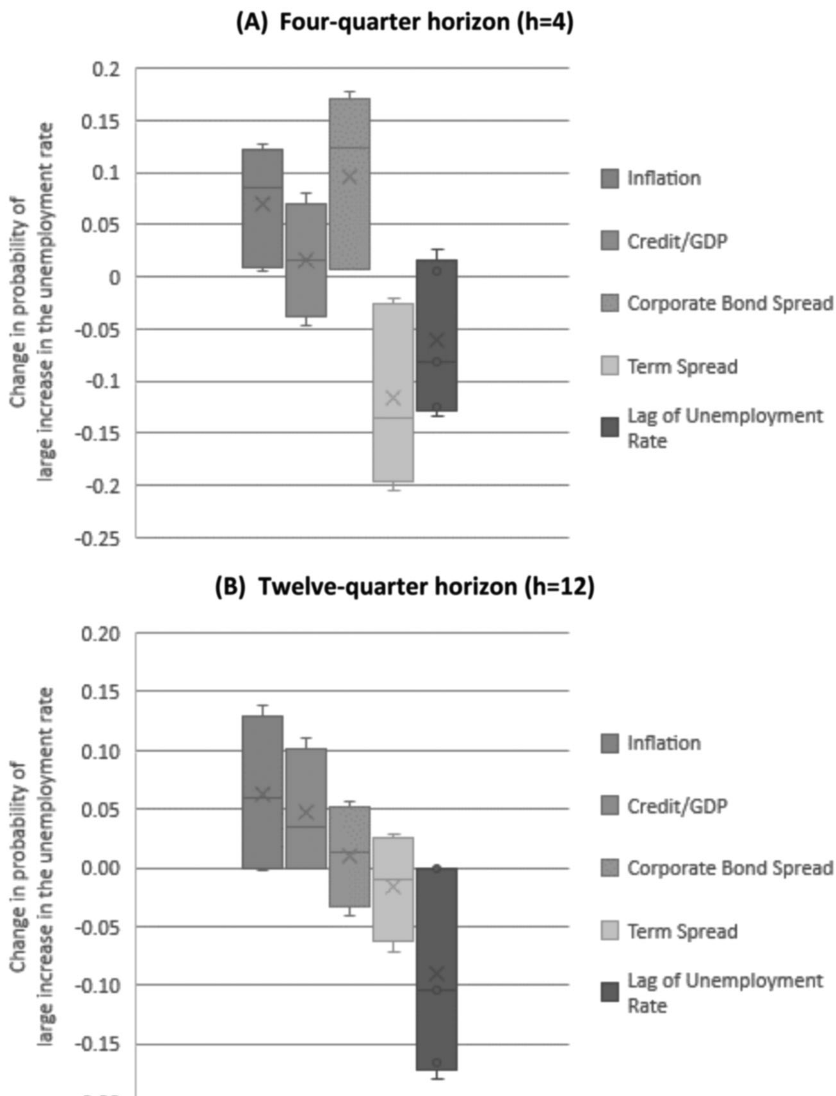

Fig 5. Marginal Effects of Variables on Probability of an Increase in the Unemployment Rate Above Its Unconditional 80th Percentile $P_{0.80}^{\Delta U}(t+h)$ (Effect of One-Standard Deviation Change in Variable). (A) Four-Quarter Horizon ($h=4$). (B) Twelve-Quarter Horizon ($h=12$).

Notes: Each graph presents statistics that summarize the marginal effects of a one-standard-deviation change in each variable on the probability of a large increase in the unemployment rate at the horizon specified (4 or 12 quarters) as implied by estimation of equation (4), evaluated at the mean value of the variables. Large increases in the unemployment rate are defined as a change in the unemployment rate from 4 or 12 quarters earlier that exceeds its unconditional 80th percentile (which equal $\frac{3}{4}$ percentage point and 1.9 percentage point, respectively). In each figure, the "x" denotes the marginal effect at the estimated parameters. The shaded box denotes the 90% confidence interval for these marginal effects. The whiskers beyond each shaded box denote the 95% confidence intervals. In all cases, confidence intervals derived via a block-of-block bootstrap using resampling of the blocks of all variables in the equation (see Kilian and Lütkepohl, 2017, chapter 12), in which the block length equals 11. Based on author's calculations using data retrieved from FRED, Federal Reserve Bank of St. Louis.

the term spread or a high corporate bond spread point to an increase in the risk of a sizable rise in the unemployment rate (as in Engstrom and Sharpe 2018). In addition, the marginal effect of inflation is sizable—high inflation points to an increase in risk of a recession, consistent with Romer and Romer (1989) and the quantile regressions. In contrast, growth in credit relative to GDP or the level of the unemployment rate have less sizable marginal effects and these effects are not statistically different than zero at the 10% level.

At the 3-year horizon, the most sizable marginal effects are those associated with inflation, credit relative to GDP, and the level of the unemployment rate. High inflation, rapid credit growth, or a low level of the unemployment rate each point to an elevated probability that the unemployment rate will increase sizably. In contrast, the marginal effects of the term spread and the corporate bond spread are not large and not statistically significant at the 10% level.

The results for the probability of an increase in the unemployment rate larger than its unconditional 80th percentile echo those from the quantile regression in Figure 3. The quantitatively large effects from the quantile regression approach to measuring unemployment risk at the 3-year horizon were those of inflation, credit growth, and the level of the unemployment rate, and the logit regressions provide a similar assessment.

3. CONCLUSIONS

The emphasis of policymakers on risk management illustrates the importance of understanding the factors that shape the risks to the economic outlook. The analysis herein has examined the relationship between unemployment risk and variables suggested in the literature on financial crises and recession prediction. Two concepts on unemployment risk were considered: the conditional 80th percentile based on a quantile regression and the probability of a large increase in the unemployment rate from a logit regression.

The results were broadly consistent. While it is challenging to estimate relationships between tail outcomes and predictors, both approaches identify important differences in the factors that appear to predict near-term and medium-term risk. Near-term risks (at a 1-year horizon) appear linked to variables identified in the recession prediction literature, including the term spread and the corporate bond spread. This finding is also similar to the emphasis on a financial conditions index in the GDP at Risk work of Adrian, Boyarchenko, and Giannone (2019). In contrast, credit and the level of the unemployment rate appear to be useful predictors of risk at a longer 3-year horizon. Note that inflation appears important at both horizons, and that relationships in the tails of distributions are challenging to estimate, yielding results that are not overwhelmingly significant in the statistical sense.

The results yield a compelling narrative. For example, the quantile regression implies large variation over time in the magnitude of unemployment risk over the medium term. The level of inflation was a large contributor to unemployment risk in the 1970s but has not been a notable contributor since the Volcker disinflation. Since the 1980s, fluctuations in credit have been important in shaping whether this measure of unemployment risk has been high or low.

The analysis of unemployment risk may provide useful information to policymakers and economists interested in understanding the magnitude and indicators of possible risks to the unemployment rate. This information may be as valuable as that in related GDP concepts, as unemployment is directly tied to the dual mandate of the Federal Reserve and has historically been a better indicator of the state of the business cycle in the United States than GDP.

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